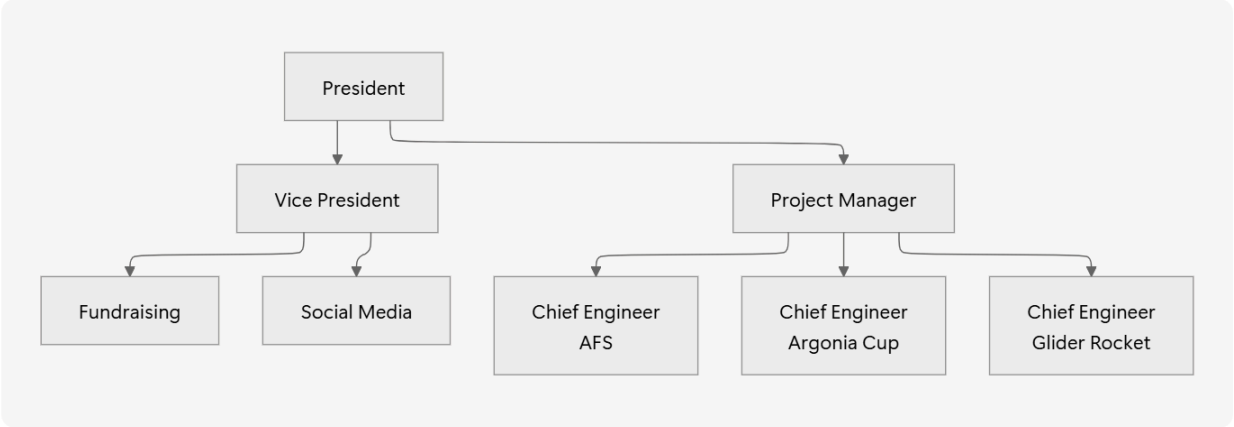
Ram Rocketry Overview

Ram Rocketry is a student-led aerospace organization at Colorado State University focused on the engineering design, analysis, and testing of high-powered rockets, with an emerging focus on propulsion systems.

Current Leadership Architecture



Our Projects

1. Active Fin Stabilization

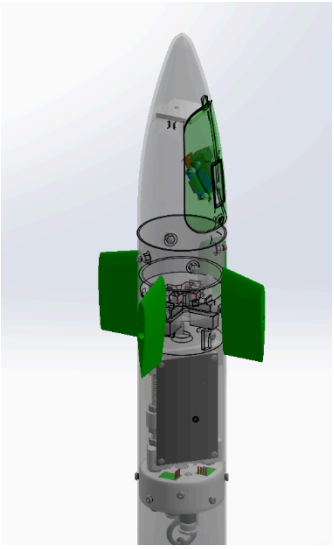
An actively controlled rocket using servo-actuated fins and onboard avionics to demonstrate 3-axis control, correct attitude in flight and optimize trajectory performance.

2. Two-Stage

A rocket designed to take the maximum number of golf balls (payload) to the highest altitude possible, while staying under a specific motor impulse range.

3. Glider Rocket

A rocket that transitions into a remotely controlled glider, using fixed wings and onboard avionics to achieve controlled, winged recovery instead of a parachute.

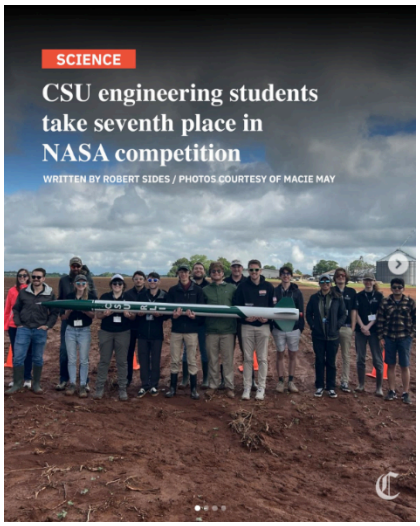


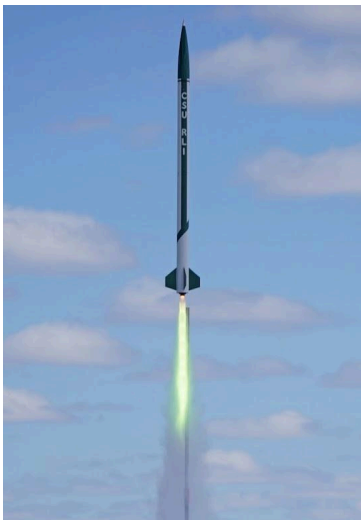


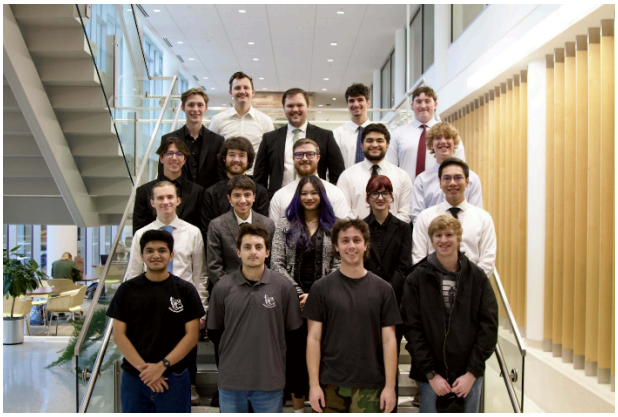


Ram Launch Initiative

RLI is CSU's only collegiate student competition team participating in the NASA University Student Launch Initiative competition (USLI) and acts as an extension of Ram Rocketry. RLI is a senior design project, and requires the core team to be all-seniors, with underclassmen interns open to volunteer and participate. Ram Rocketry acts as a feeder club into RLI.







Liquid Program

Our vision is to establish the first student-led liquid rocket engine program at CSU, focused on rigorous engineering, safe testing practices, and industry-relevant design workflows.

Our goal is to learn fast, build the engine correctly, and execute safely.

Engine Characteristics (Open to feedback and amends):

- Traditionally CNC Manufactured Stainless Steel
- Thrust: 1100N
- F: Diluted Ethanol (80% C2H5OH + 20% H2O)
- O: Nitrous Oxide (N2O)
- Ablative Cooling
- Pintle Injector

